

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).*

Assessment Tool Weightage: 25%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Data Interpretation

Duration: 1hr

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.
2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance
3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results.

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance.

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data.

ANSWERS:

1. Reward values increase gradually from 10 to 45

Relevant RL Concept

Learning Performance and Policy Improvement

Analysis

The reward increases steadily from 10 to 45 over multiple episodes. This indicates that the agent is learning from its interactions with the environment and improving its decision-making. Initially, the agent explores different actions, resulting in lower rewards. As training progresses, it identifies better actions and updates its policy.

Interpretation

- Increasing rewards indicate successful learning.
- The policy becomes more effective over time.
- The agent makes fewer poor decisions.
- The cumulative reward increases because the agent selects actions that maximize long-term rewards.

Conclusion

The increasing reward trend shows that the RL agent is learning successfully, improving its policy, and converging toward an optimal solution.

2. ϵ -Greedy Agent ($\epsilon = 0.1, 0.3, 0.5$)

Relevant RL Concept

Exploration vs. Exploitation

Observation

ϵ Value	Cumulative Reward
0.1	Highest
0.3	Medium
0.5	Lowest

Analysis

A smaller ϵ value (0.1) means the agent exploits the best-known action most of the time while still exploring occasionally. A larger ϵ value (0.5) causes frequent random exploration, reducing the opportunity to exploit learned knowledge.

Interpretation

- $\epsilon = 0.1$: Better balance, highest reward.
- $\epsilon = 0.3$: Moderate exploration and exploitation.
- $\epsilon = 0.5$: Excessive exploration, lower reward.

Conclusion

The agent performs best with $\epsilon = 0.1$, as it effectively balances exploration and exploitation, resulting in the highest cumulative reward.

3. Comparison of Q-Learning and SARSA

Relevant RL Concept

Q-Learning vs. SARSA (Off-policy vs. On-policy)

Observation

Q-Learning achieves slightly higher rewards than SARSA over 100, 200, and 300 episodes.

Analysis

Q-Learning is an off-policy algorithm that updates values using the maximum possible future reward, leading to faster convergence. SARSA is an on-policy algorithm that updates values based on the action actually taken, making it more conservative.

Interpretation

Algorithm	Performance
Q-Learning	Higher rewards
SARSA	Slightly lower rewards

Conclusion

Q-Learning converges faster and achieves higher average rewards, while SARSA provides safer and more conservative learning.

4. Discount Factor ($\gamma = 0.2, 0.5, 0.9$)

Relevant RL Concept

Discount Factor and Long-Term Reward Optimization

Observation

Discount Factor	Average Reward
0.2	Low
0.5	Medium

Discount Factor	Average Reward
0.9	High

Analysis

The discount factor determines the importance of future rewards.

- $\gamma = 0.2$: Focuses mainly on immediate rewards.
- $\gamma = 0.5$: Balances immediate and future rewards.
- $\gamma = 0.9$: Gives high importance to future rewards, encouraging long-term planning.

Interpretation

As γ increases, the agent considers long-term benefits more effectively, leading to improved average rewards.

Conclusion

A discount factor of 0.9 is most suitable because it encourages long-term reward optimization and results in better learning performance.

5. Success Rate Increases from 45% to 92%

Relevant RL Concept

Policy Learning and Convergence

Observation

Episodes	Success Rate
50	45%
200	92%

Analysis

The steady increase in success rate indicates that the agent is continuously improving its policy through repeated interactions with the environment. Higher success rates reflect better action selection and fewer mistakes.

Interpretation

- The agent learns from experience.
- Policy quality improves over time.
- Performance becomes more stable.
- A 92% success rate suggests the policy is close to optimal.

Conclusion

The increasing success rate demonstrates effective learning and indicates that the agent has nearly converged to an optimal policy.

Code of Conduct:

*I **Kandukuri.Kavya-192425283** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Kandukuri.Kavya-192425283

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand